

Drugs Acting on Gonadal Function

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Department of Pharmacology

- ❖ **성호르몬 분비 조절 기전을 이해한다.**
- ❖ **성선기능 조절의 기본 원리를 이해한다.**
- ❖ **성선기능 조절약물의 종류를 알고 약리학적 기전을 이해한다.**

Androgens (19 carbon)

Cholesterol

Dehydroepiandrosterone

16α-Hydroxyandrostenedione

Androstenedione

Testosterone

Dihydrotestosterone

Estrogens (18 carbon)

Estrone

Estradiol

Estriol

Biological Activity

Estradiol > Estrone > Estriol

Progestagens (21 carbon)

Pregnenolone

Progesterone

17α-Hydroxypregnenolone

17α-Hydroxyprogesterone

11-Deoxycorticosterone

Corticosterone

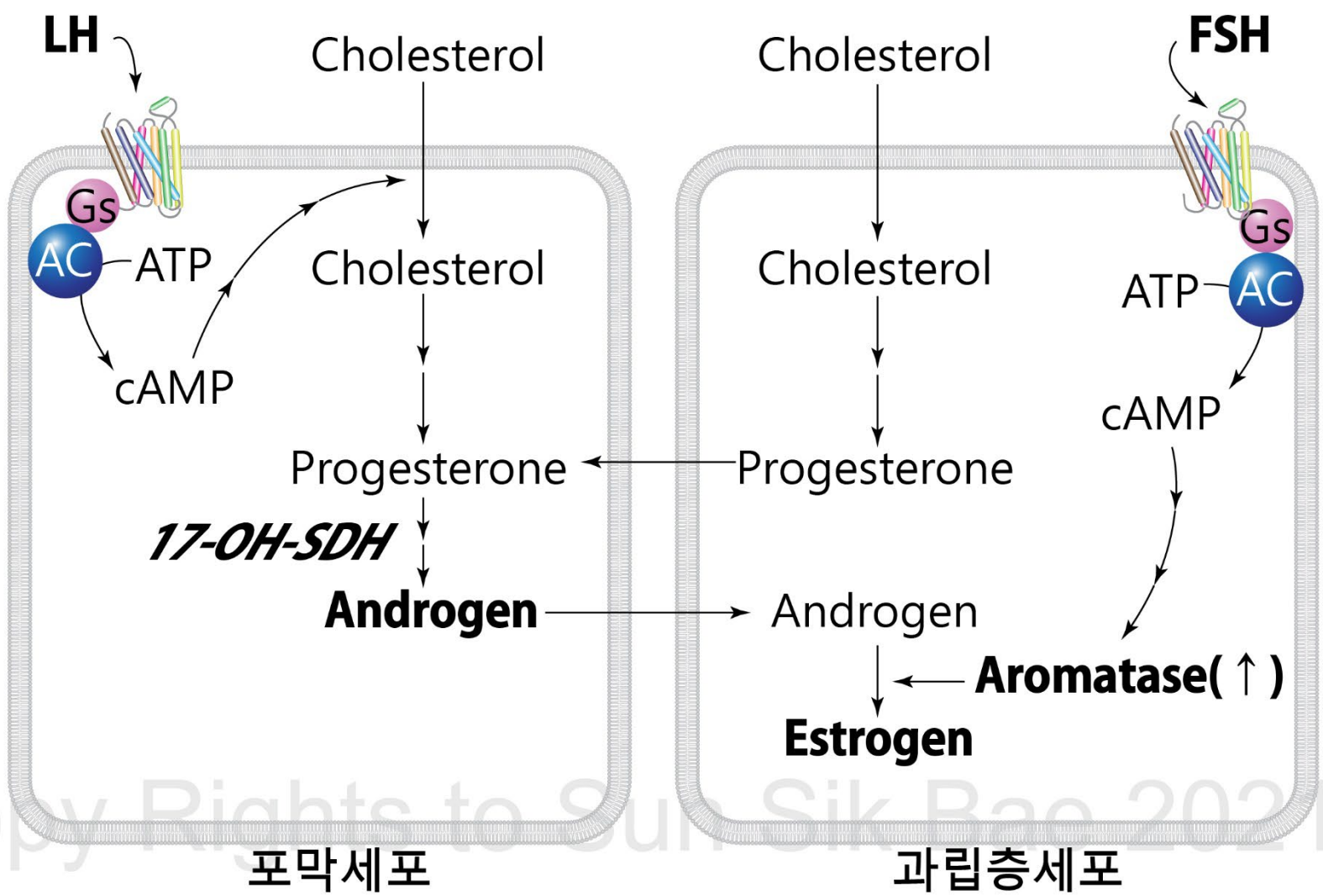
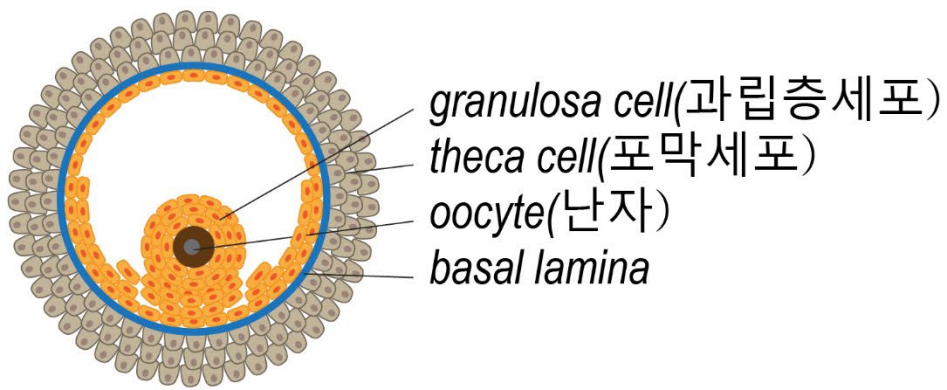
11-Deoxycortisol

Cortisol

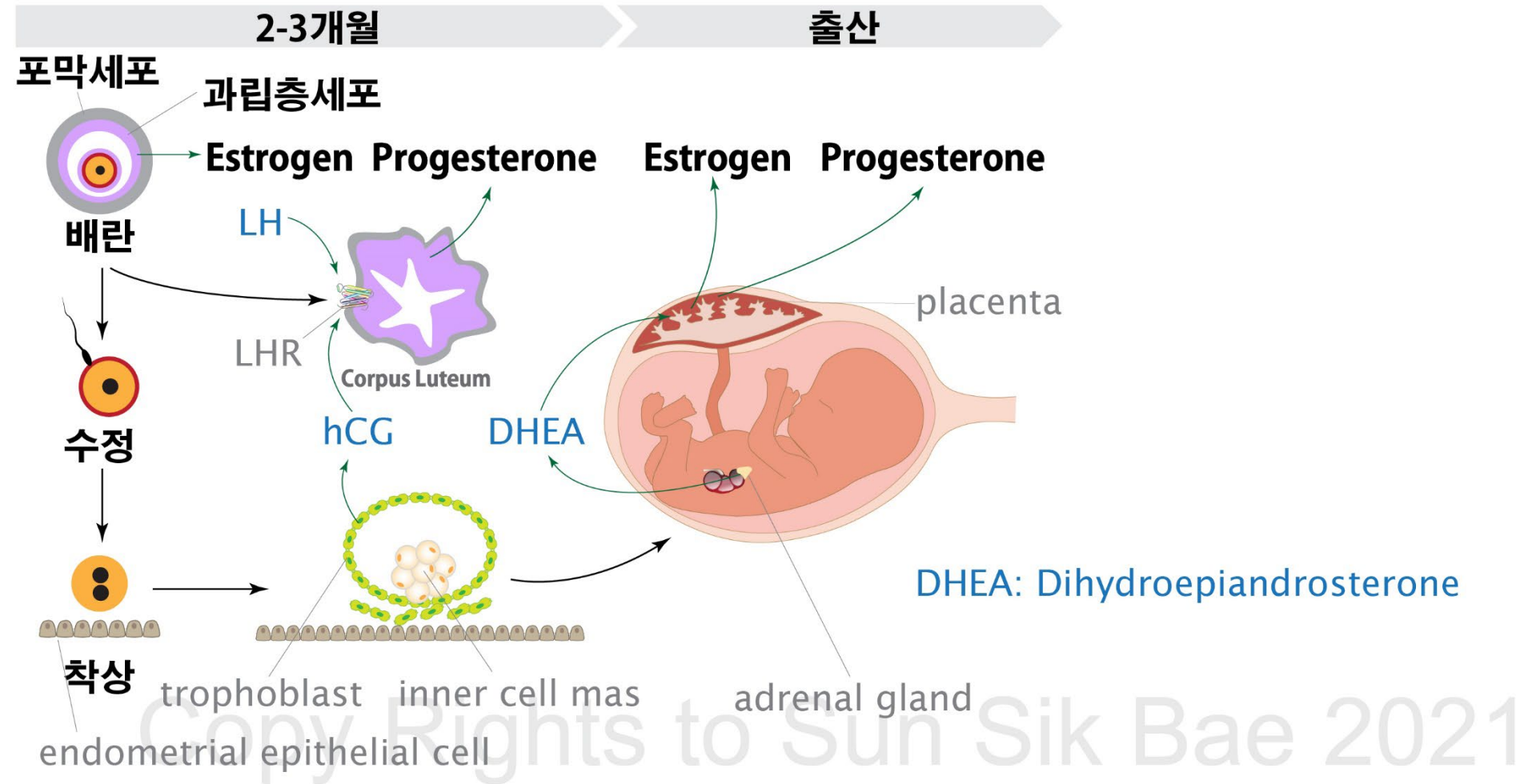
18-Hydroxycorticosterone

Aldosterone

► Estrogen synthesis



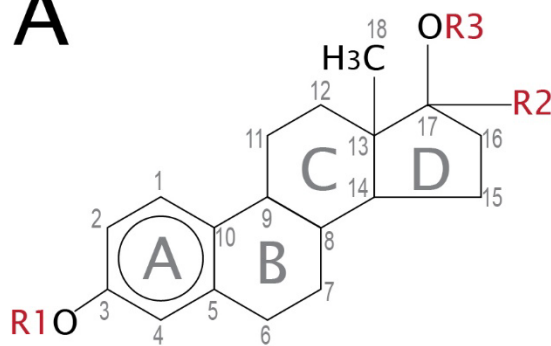
► Progesterone synthesis



► Estrogen and Progestins

Estrogens

A



Steroidal Backbone

[Steroidal]

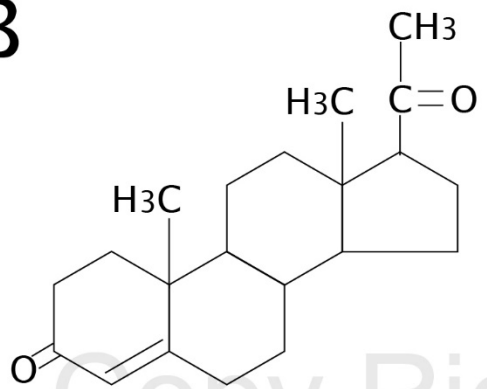
- > Estradiol
- > Estradiol valerate
- > Ethinyl estradiol
- > Mestranol
- > Estrone sulfate
- > Equilin

[Nonsteroidal]

- > Diethylstilbesterol
- > Bisphenol A
- > Genistein

Progestins

B



Progesterone

[Pregnanes]

- > Progesterone
- > Medroxyprogesterone acetate

[Estranes]

- > 19-nortestosterone
- > Norethindrone

[Gonanes]

- > Norgestrel
- > Norgestimate

► Regulation of development

Estrogen

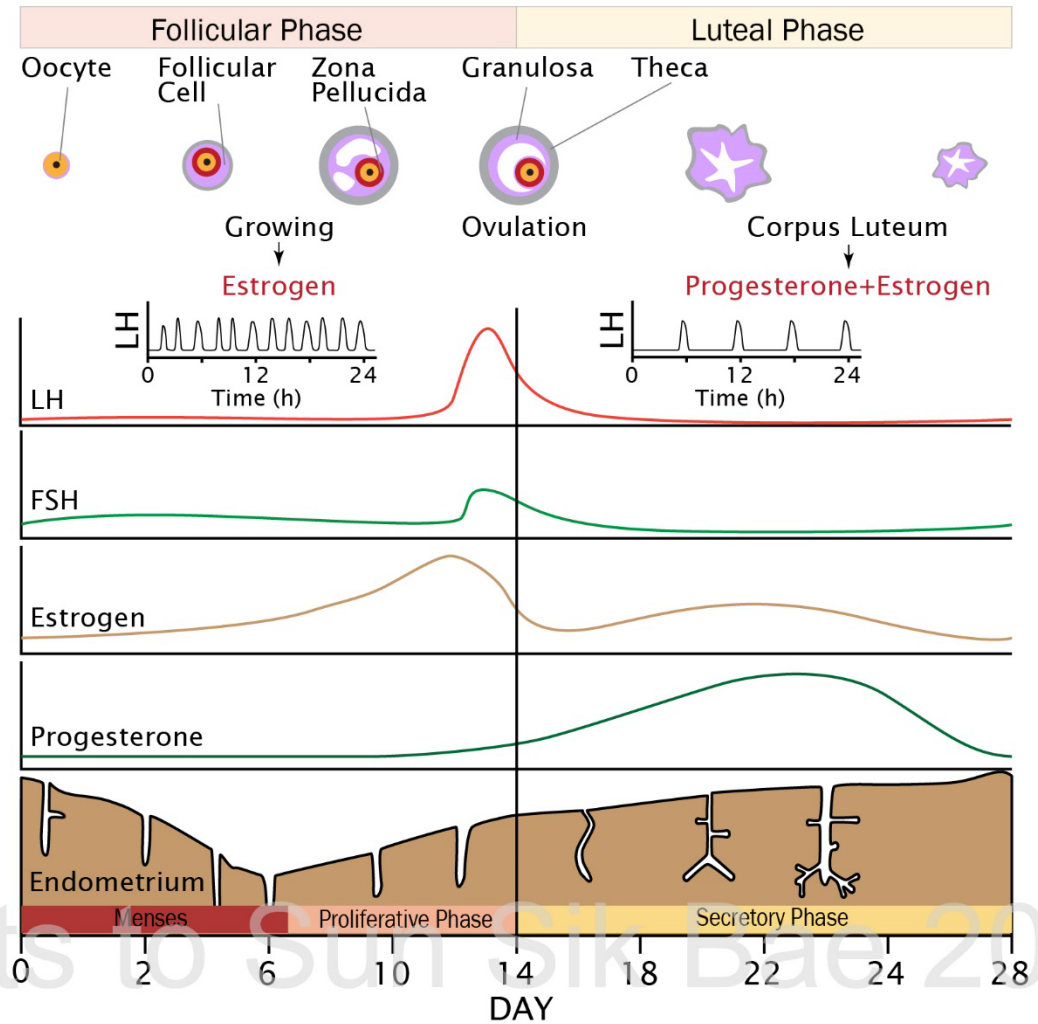
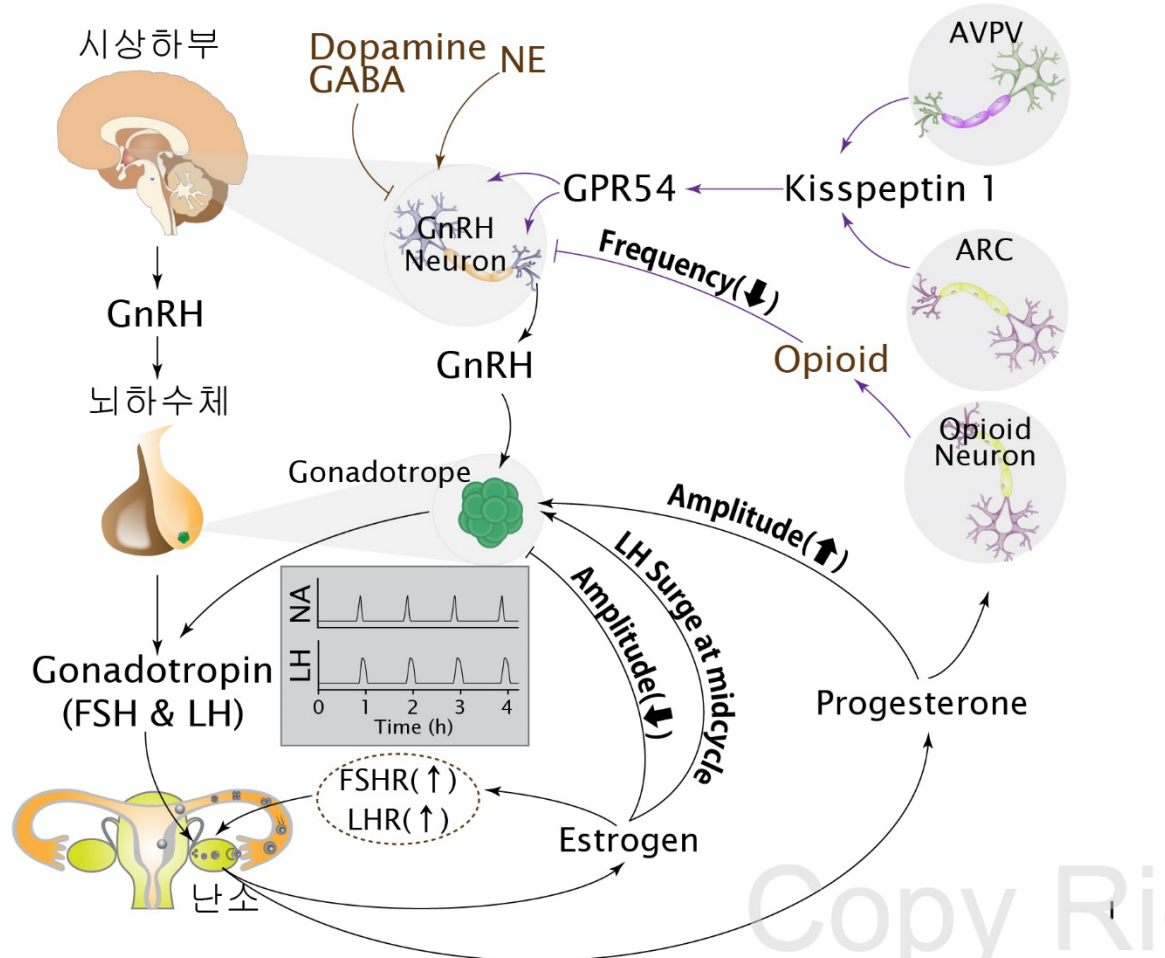
- > Pubertal changes in girls
- > Growth and Development of
: Vagina, Uterus, Fallopian tube, Breast enlargement
- > Body contour
- > Pubertal long bone growth spurt and epiphyseal closure

Estrogen in male

- > Estrogen deficiency in **boy**:
 - » Diminished pubertal growth spurt
 - » Delayed skeletal maturation
 - » Delayed epiphyseal closure-->Linear growth
- > Estrogen deficiency in **adult**
 - » Elevated gonadotropin
 - » Macroorchidism
 - » Abnormal carbohydrate and lipid metabolism

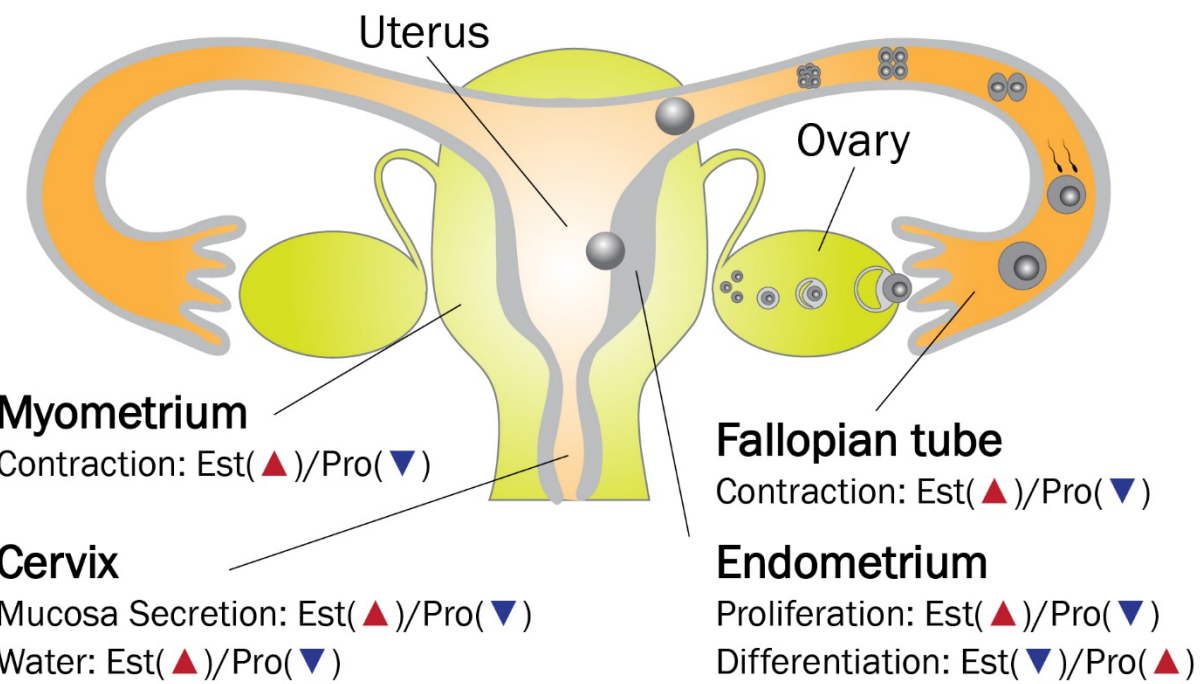
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► Regulation of menstrual cycle



AVPV:Anteroventral periventricular nucleus; ARC: Arcuate nucleus; NA: Neuronal Activity

► Regulation of reproductive organ



Est: estrogen; Pro: progesterone

Endometrium Composition

- Epithelial cells
- Stroma (Blood vessels, fibroblasts, macrophage, lymphocyte)

Proliferation stage:

- Estrogen: Rebulid endometrium by the proliferation and differentiation
- Estrogen induces progesterone receptor (PR), enable cells to respond progesterone

Luteal stage:

- Progesterone **limits the proliferative effect of estrogen**
- Progesterone provides “epithelial secretion” for implantation and blood vessel formation

Implantation:

- Induces secretion of **chorionic gonadotropin (hCG)** from trophoblast
- **hCG** activates LH receptor on corpus luteum to **maintain steroid hormone synthesis**
- hCG is produced from **placenta** at later stage of pregnancy

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► Regulation of metabolism

Bone

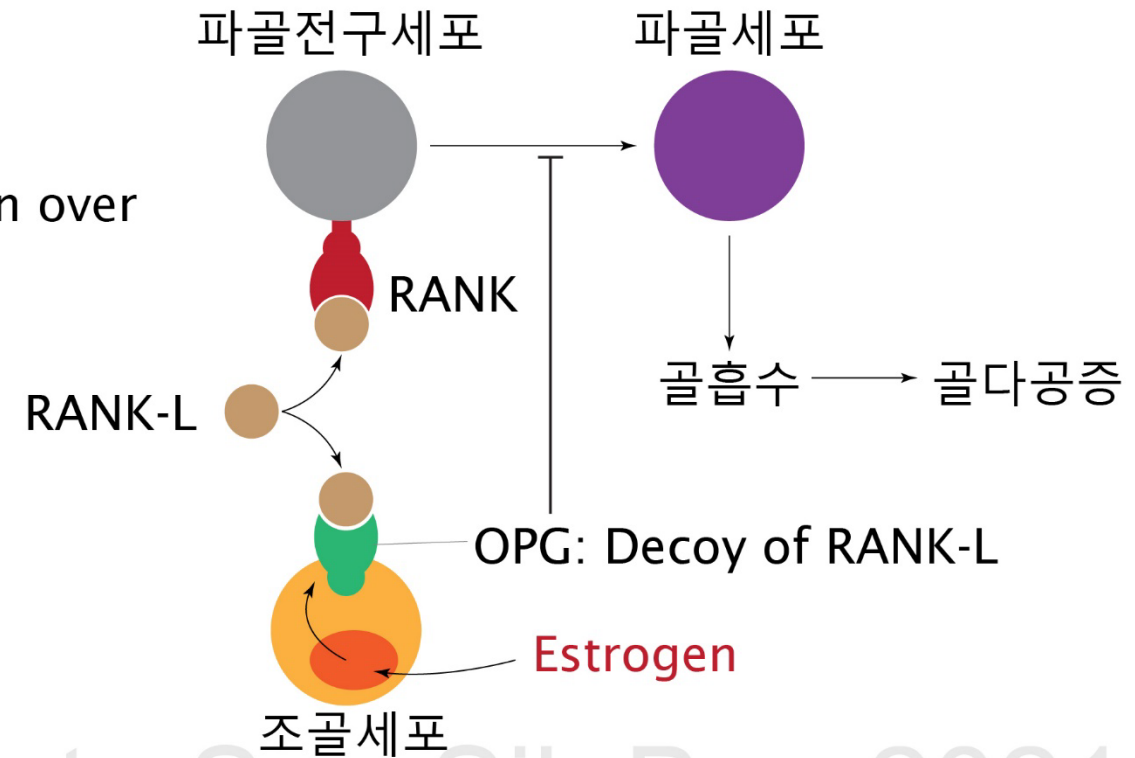
- > Positive effect on bone mass
 - » Osteoclast(↓), Osteoblast(↑)
- > ER defective man
 - » Osteoporosis, unfused epiphyses, increased bone turn over

Lipid

- > Estrogen Effects:
 - » TG(↑), Cholesterol(↓), LDL(↓), HDL(↑)
 - » Beneficial role in LDL/HDL ratio

Vasculature

- > Estrogen Effects:
 - » NO(↑), Atherogenesis(↓)
 - » EC growth(↑), SMC growth(↓)



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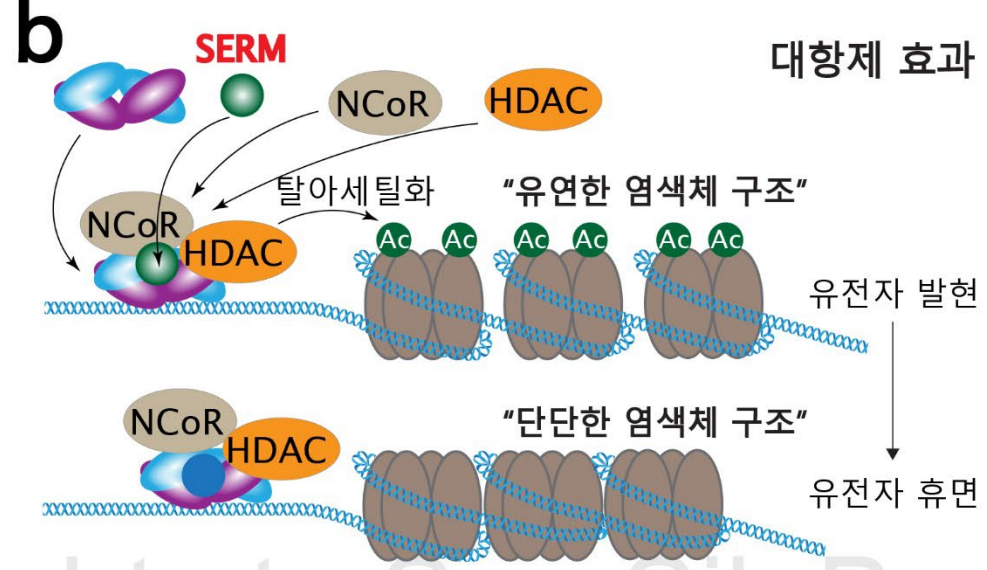
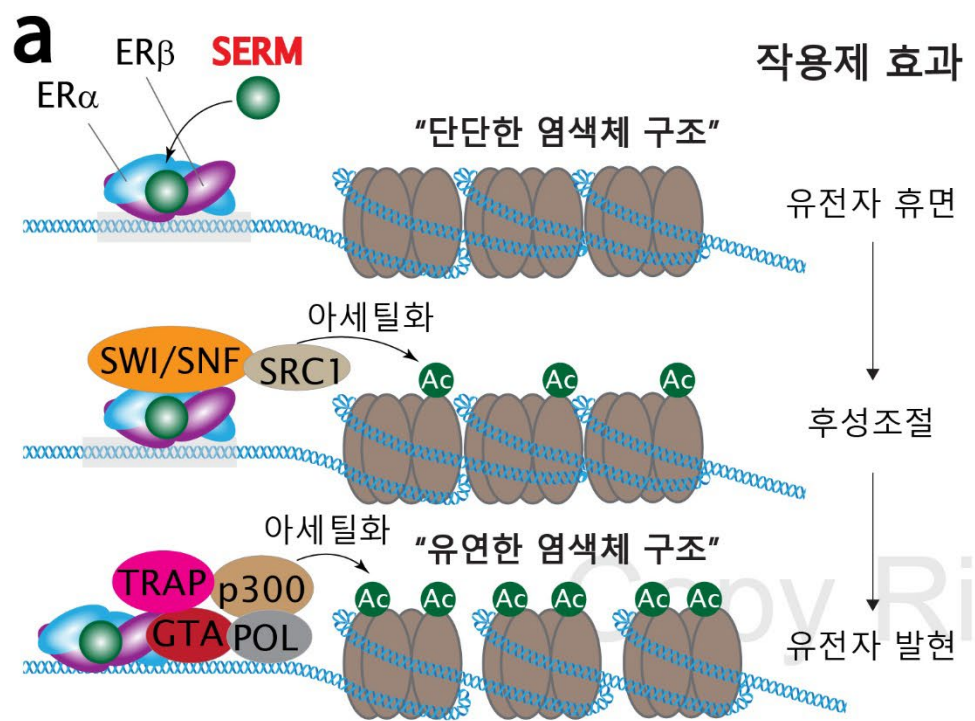
► Selective Estrogen Receptor Modulator (SERM)

Definition

> Transcriptional inhibition varies from tissues and cells

Mechanism of action

> Alters transcriptional activity of estrogen receptor

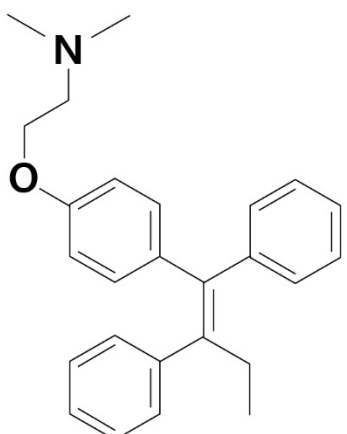


► Antiestrogen

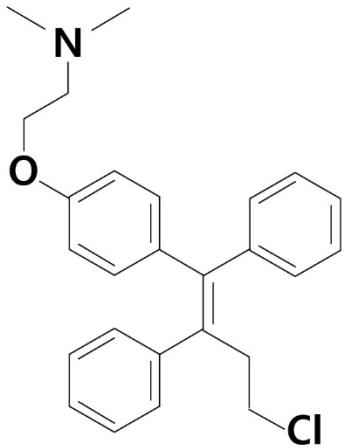
Definition

> Inhibits estrogen effect in all tissues

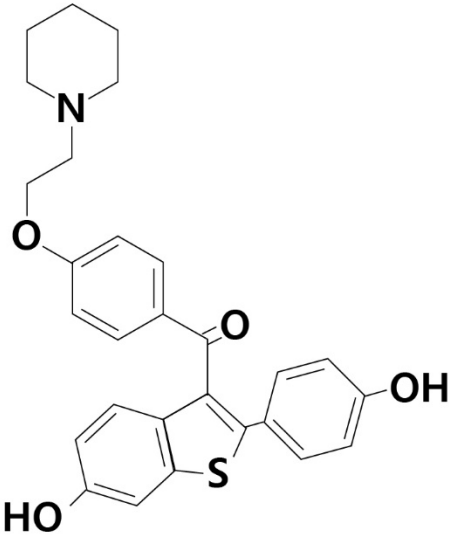
SERM



Tamoxifen

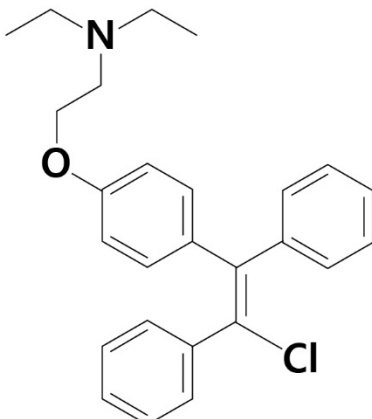


Toremifene

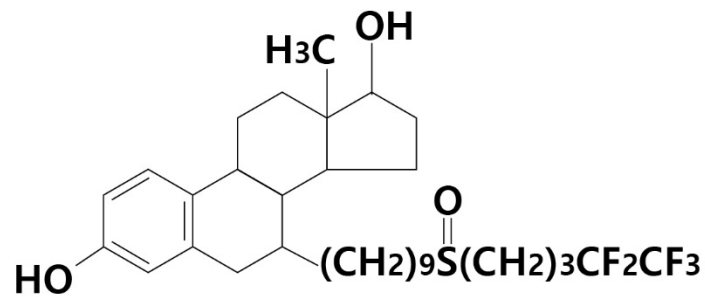


Raloxifene

Antiestrogen



Clomiphene



Fulvestrant

Tamoxifen: Antagonistic effect in breast tissues and osteocytes; agonistic effect endometrium
Raloxifen: Agonistic effect in osteocytes; antagonistic effect in other tissues

▶ Pharmacology and clinical application

[유방암]

Tamoxifen: ER⁺ 유방암의 일차 치료약물 (50% 반응성)

Toremifene: 치료효과는 tamoxifen과 유사함

Fulvestrant: tamoxifen에 내성이 생긴 유방암의 이차 치료약물

[골다공증]

Raloxifene: 척추 골밀도를 2% 증가시킴

[불임증]

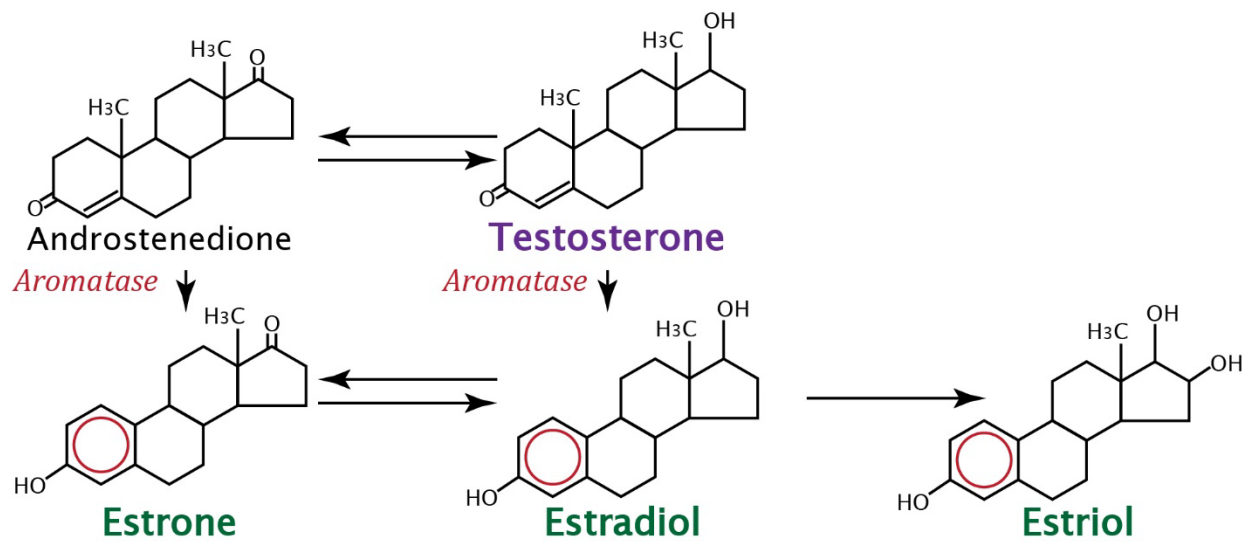
Clomiphene: FSH 분비를 촉진해서 배란 촉진

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► Pharmacology and clinical application

> Target key enzyme in estrogen synthesis

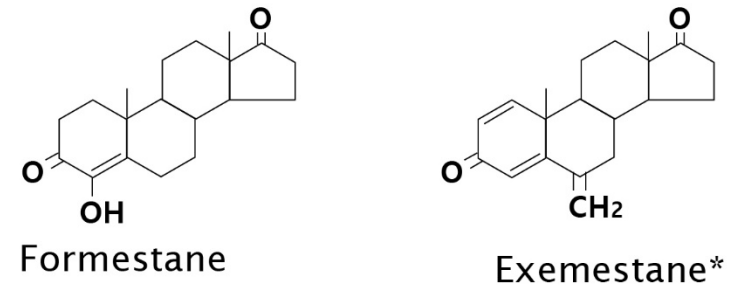
» Aromatase plays an essential role



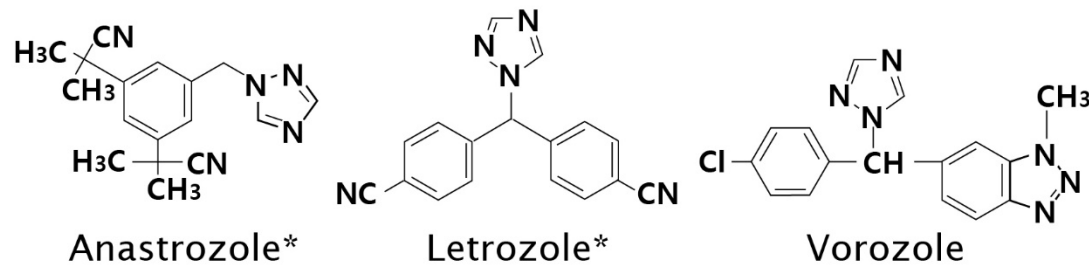
> Applications

- » First-line treatment of breast cancer
- » Second-line drug after tamoxifen

Steroidal Aromatase Inhibitor(비가역적)

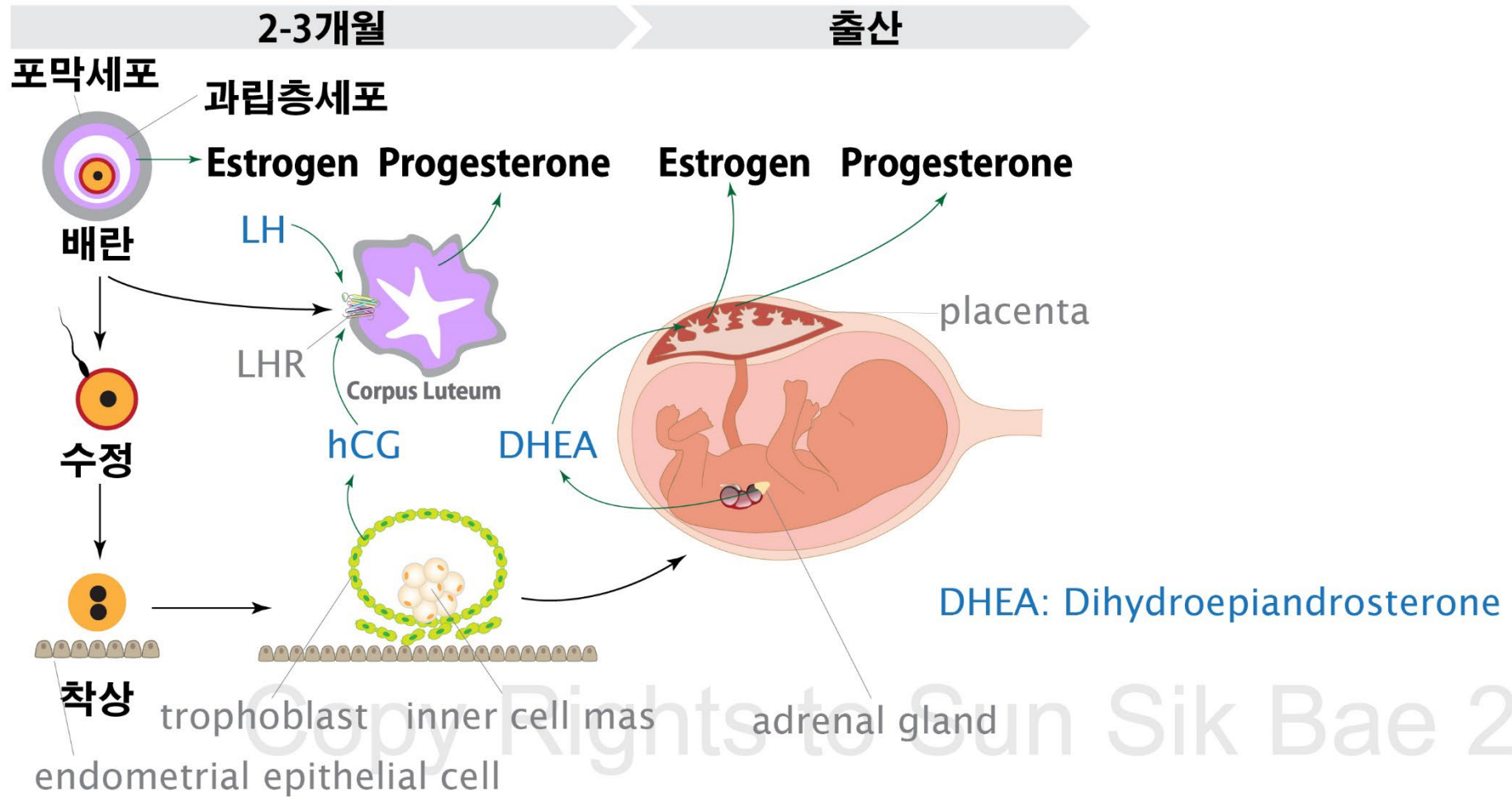


Non-steroidal Aromatase Inhibitor(가역적)



*U.S. approved

Biosynthesis



► Physiological action

> Neuroendocrine action

- » Decrease frequency of GnRH pulse
: major mechanism of **contraceptive**

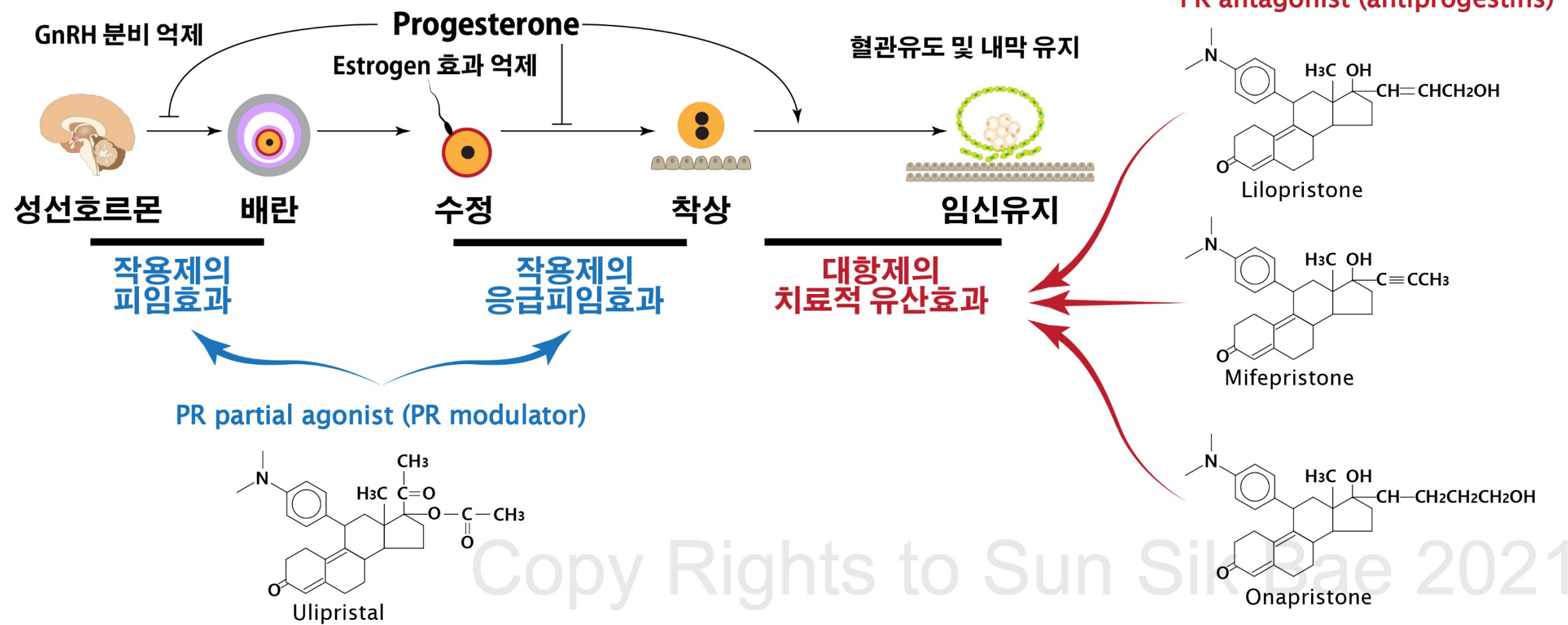
> Reproductive tract

- » **Block** estrogen-dependent endometrium **proliferation**
- » Maintain secretory endometrium
- » Dropping progesterone initiates menstruation
- » Block estrogen-induced mucus and water secretion at cervix
: **Block sperm penetration**
- » Maintain pregnancy
- » Suppress menstruation and **uterine contractility**

> Mammary gland

- » Induce proliferation of **acini** of the mammary gland together with estrogen (at pregnancy stage)
- » Induce single round of proliferation of mammary epithelial cells (at normal cycle)

► Pharmacology of PR modulator and antiprogestins



Progesterone receptor regulator

► Clinical application of antiprogestins

> Antiprogestine effect

> Used for **termination of pregnancy**

> Pharmacological effects

» Competitive PR antagonist (Onapristone, Mifepristone)

» Mechanism of action

: Break down PRs--> Induce detachment of blastocyst-->Lower hCG-->**Lower Progesterone**
-->Can't maintain pregnancy

: **Lower Progesterone**--> Increase prostaglandin synthesis--> Induce contraction of uterus
--> Expulsion of detached blastocyst

» **Binds to glucocorticoid and androgen receptor**

: Block negative feedback loop of cortisol thereby **enhance ACTH** (pituitary)
--> Increase glucocorticoid and mineralocorticoid hormones

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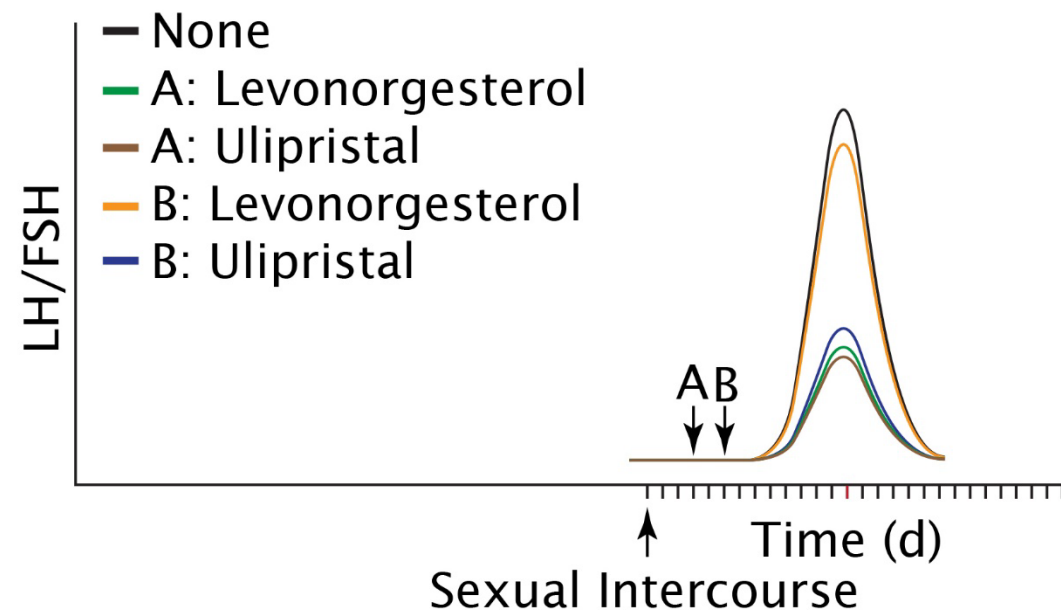
Progesterone receptor regulator

► Clinical application of PR modulator

- > Selective PR modulator (Partial agonist of PR)
- > Used for **inhibition of ovulation (emergency contraceptive)**
- > Pharmacological effects
 - » Behave like progesterone
 - » Inhibition of LH release
 - » Inhibition of LH-induced follicular rupture
 - » Comparison with progestin (e.g., levonorgestrel)

Effective times after unprotected sexual intercourse

Ulipristal	<120 h
Levonorgestrel	<72 h



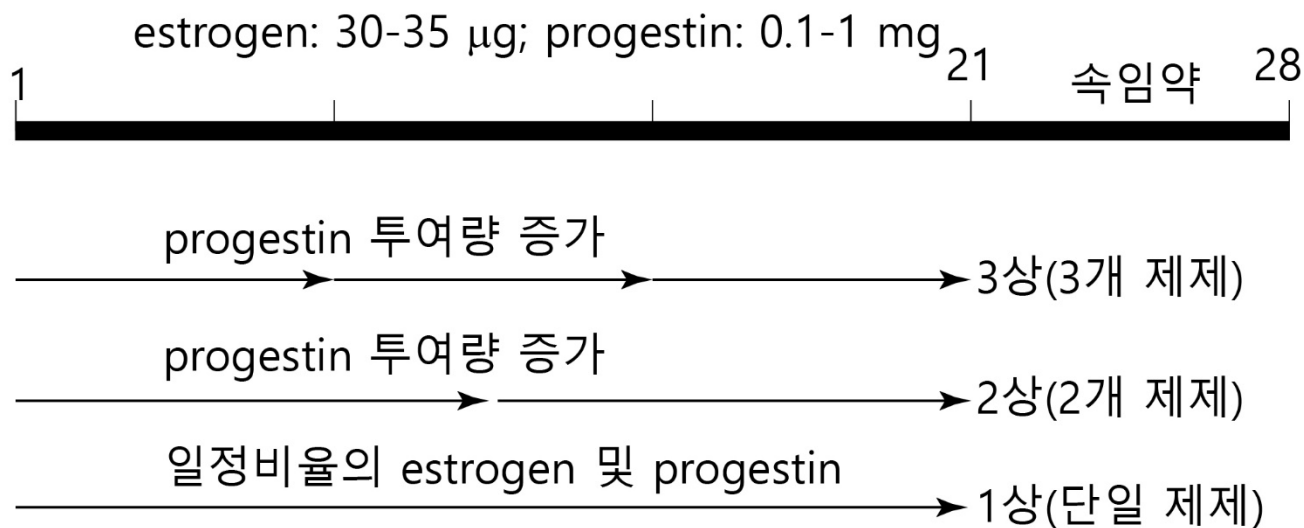
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Therapeutic uses of estrogen and progestins

► Hormonal contraception

> Combination oral contraceptives

- » Contains both estrogen (ethylestradiol) and progestin (19-nortestosterone)
- » Efficacy > 99.9%
- » Protocol (daily)



» Mechanism

: LH and FSH level down, no midcycle LH surge-->Block ovulation

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► Hormonal contraception

> Progestine-only contraceptive

- » Efficacy >99%
- » Possible for subdermal implant (norgesterol > 5 yrs; etonorgesterol > 3 yrs)
- » Mechanism
: Thickening cervical mucus--> **block sperm penetration**

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► Postcoital (Emergency) contraception

> Mechanism of action

- » Block ovulation
- » Affects gamet function and survival (egg or sperm)
- » Block proper implantation

> Ulipristal (Mifepristone is not approved in U.S.)

- » Selective progesterone receptor modulator (PRM)
- » Up to 120 h after unprotected sexual intercourse

> Plan B (levonorgestrel)

- » Option I: Take 0.75 mg of each within 72 h of intercourse and followed by second dose 12 h later
- » Option II: Take 1.5 mg of each within 72 h of intercourse

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▶ Termination of pregnancy

> Mifepristone (600 mg)

» Antiprogesterone

» Take within 49 days of last menstruation followed by misoprostol (PGE₂, 400mg) 48 h later

▶ Induction of sexual maturation

> Turner Syndrome (Only showed in female by missing one of the X chromosomes)

» Estrogen is usually used

▶ Induction of ovulation

> Unovulation accounts for 50% of female infertility

> Treatment principle

» Block negative feedback inhibition of FSH/LH production

> Type of agents

» Clomiphene (antiestrogen)

» Aromatase inhibitors (e.g., letrozole, vorozole)

» Menotropin (Human Menopausal Gonadotropin, hMG)

Urine extracts from postmenopausal women have high level of FSH and LH

► Postmenopausal Treatment

> Postmenopause

- » No menses within 1 year
- » In between 45-60 yrs
- » Adverse effects were mainly caused by low level of estrogen resulted from no follicular activity in ovary

> Symptoms

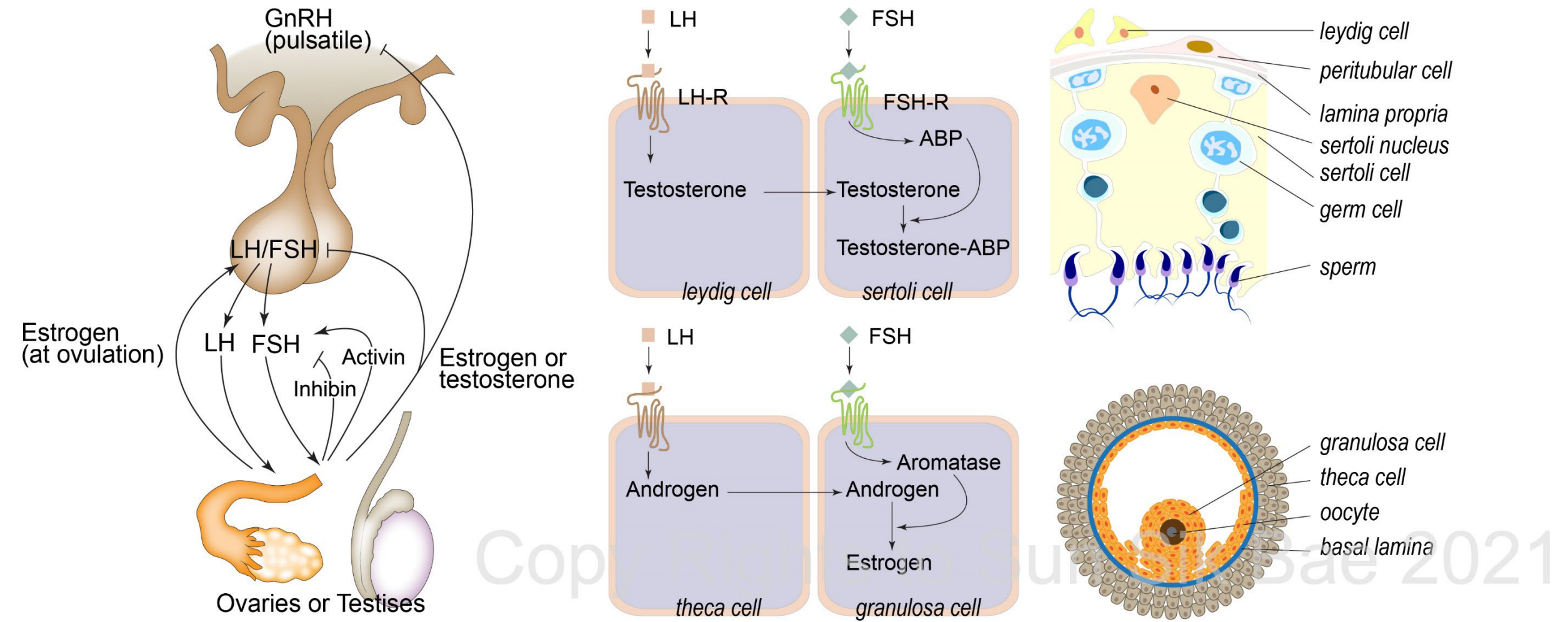
- » Hot flush, sweat, insomnia, osteoporosis, cognitive disorder, amnesia(기억력 감퇴)

> Treatment

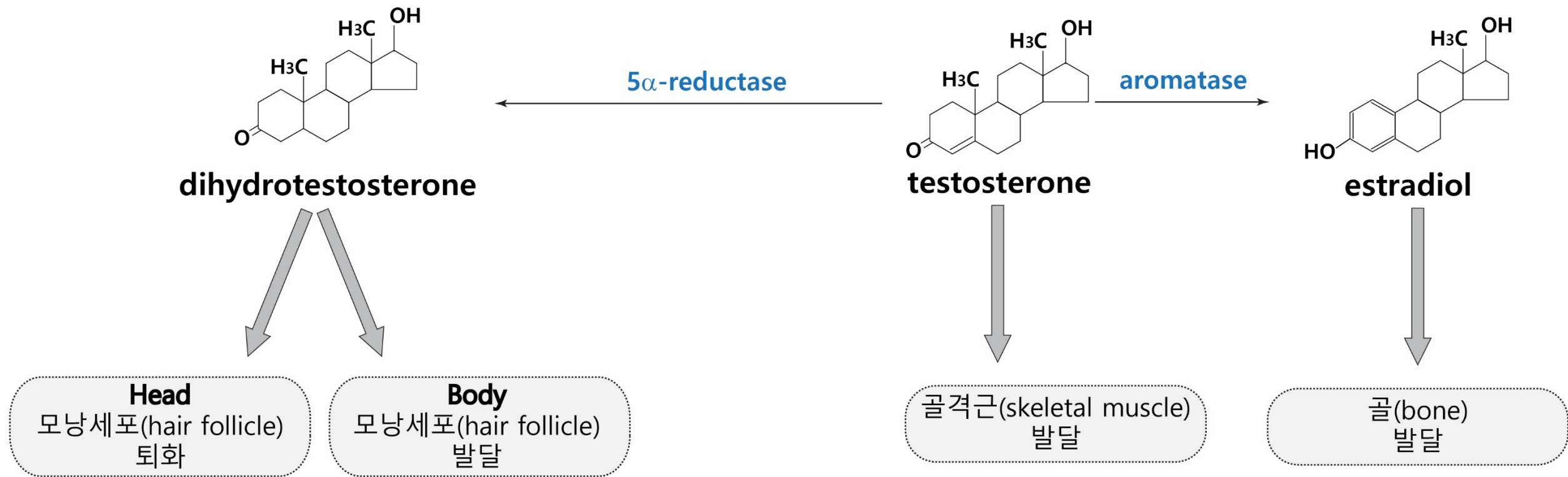
- » Equine estrogen(0.625mg/d) or ethynyl estradiol(20-35ug/d)
- » Progestins(norethindrone, norgestrel, levonorgestrel) can be used to block endometriosis which is caused by estrogen-induced proliferation of epithelial cells

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► Mechanism of androgen synthesis

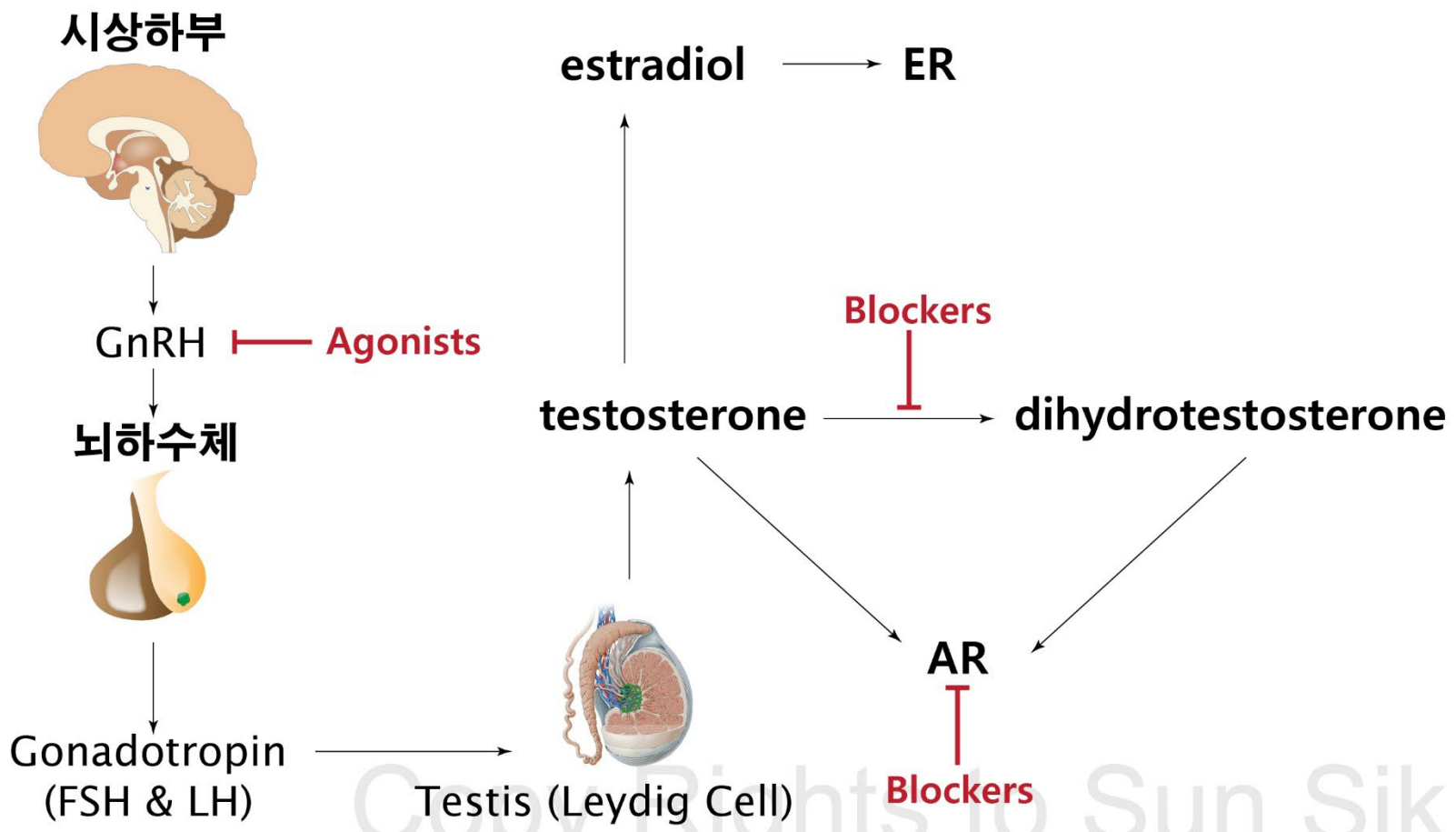


► Male sex hormone metabolites and biological action



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► Site of action

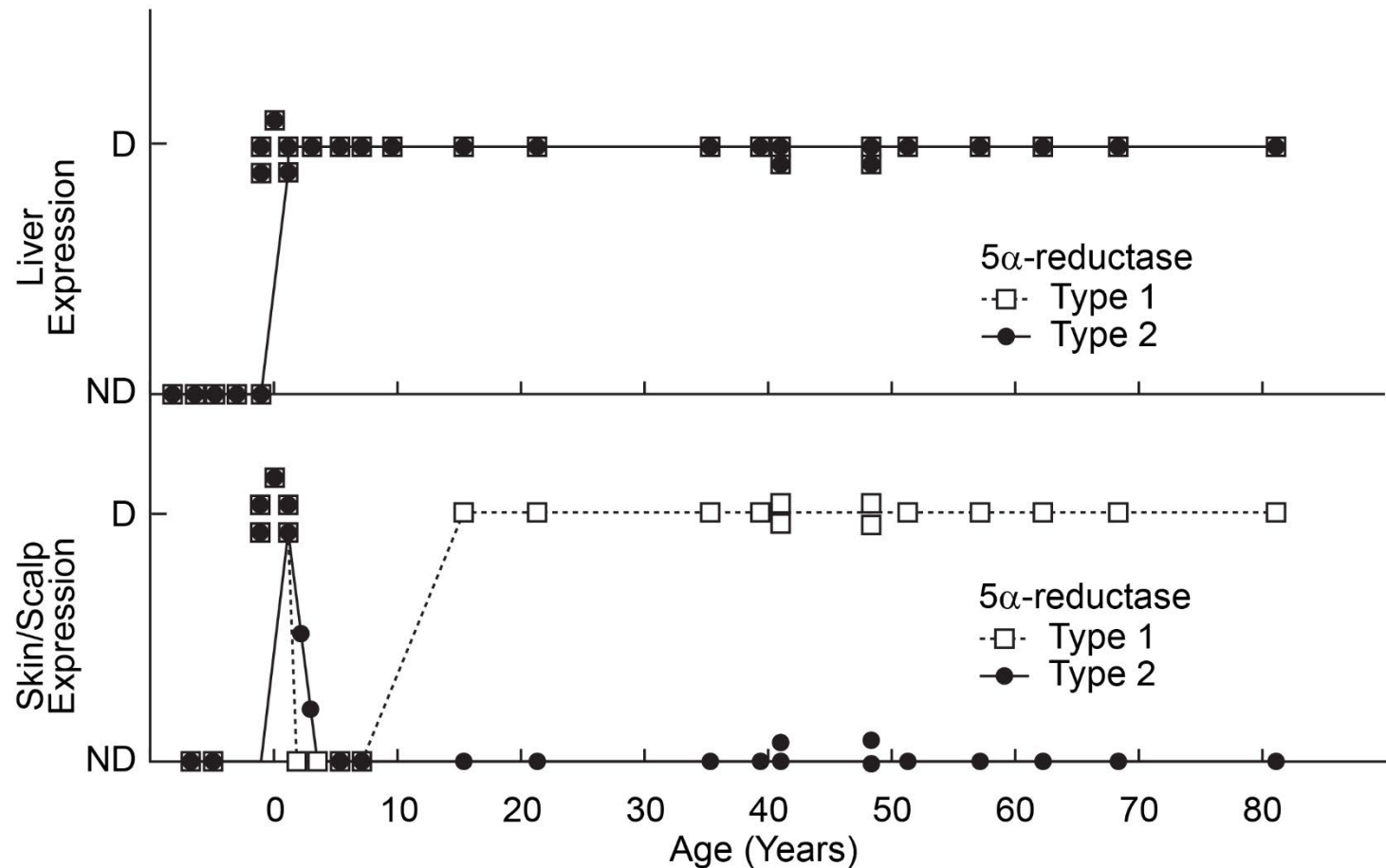


► Therapeutic application

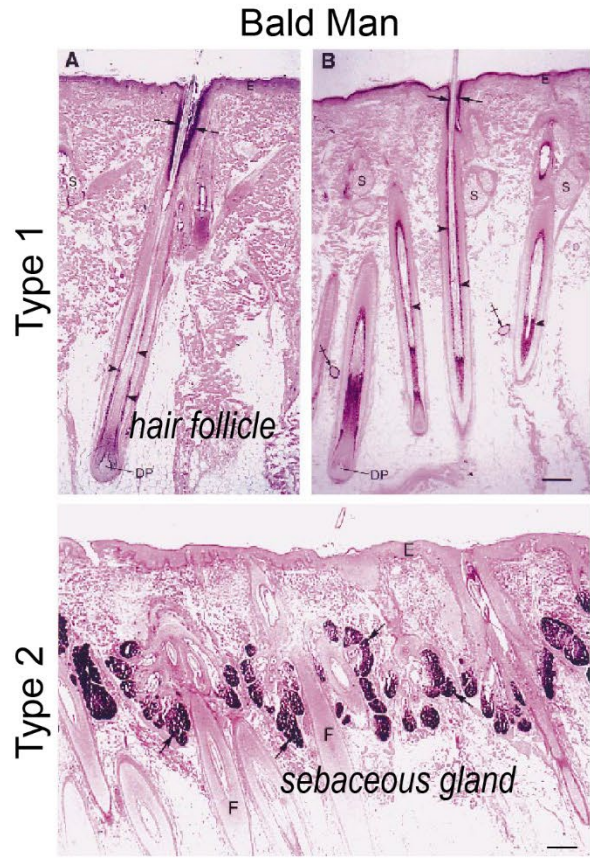
Drugs	Therapeutic Uses
GnRH Receptor Agonists	
Leuprolide	Metastatic Prostate Cancer Endometriosis(Leuprolide) Breast Cancer(Goserelin)
Goserelin	
Triptorelin	
Histrelin	
Buserelin	
AR Antagonists	
Flutamide	Metastatic Prostate Cancer
Bicalutamide	
Nilutamide	
Enzalutamide	
5α-Reductase Inhibitors	
Finasteride	Benign Prostate Hyperplasia Alopecia(Finasteride)
Dutasteride	

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► Expression pattern of 5α-reductase



[JCI, 1993, Vol 92, 903-910]



[Br. J. Derm., 1999, Vol 141, 481-491]

► Side effect of finasteride



Drug Design, Development and Therapy

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ORIGINAL RESEARCH

Protective effect of DA-9401 in finasteride-induced apoptosis in rat testis: inositol requiring kinase I and c-Jun N-terminal kinase pathway

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Yu Seob Shin^{1,*}
Bo Ram Choi¹
Keshab Kumar Karna¹
Hye Kyung Kim²
Sung Woon Lee³
Chul Young Kim⁴
Jong Kwan Park¹

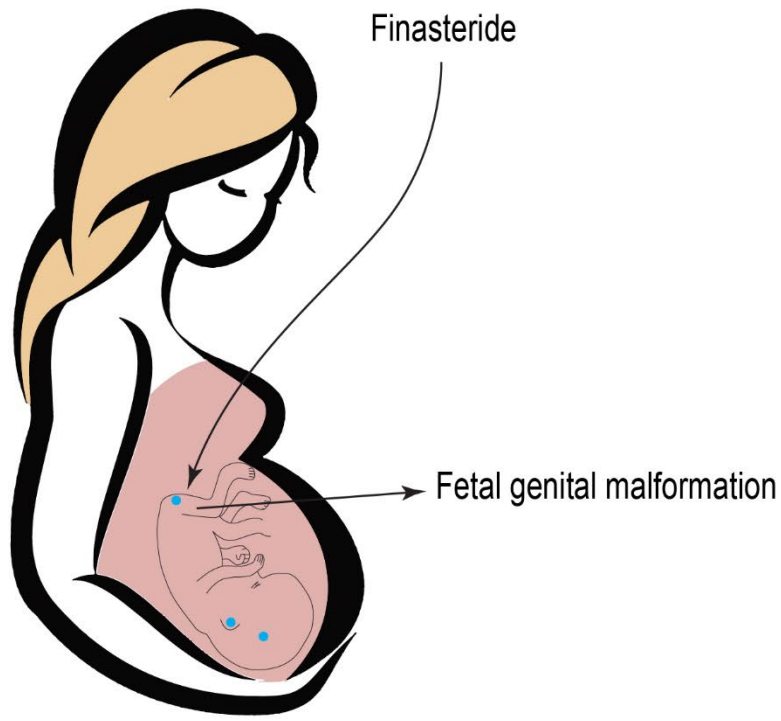
¹Department of Urology, Chonbuk National University and Research Institute of Clinical Medicine of Chonbuk National University, Biomedical Research Institute and Clinical Trial Center of Medical Device of Chonbuk National University, Jeonju, ²College of Pharmacy, Kyungsu University, Busan, ³Department of Urology, Samsung Medical Center, Samsung Biomedical Research Institute, Sungkyunkwan University Medical School, Seoul, ⁴College of Pharmacy, Hanyang University, Ansan, Republic of Korea

Abstract: Finasteride is used to treat male pattern baldness and benign prostatic hyperplasia. This study investigated the toxicity of finasteride and recovery by DA-9401 using Sprague Dawley (SD) rats. Forty adult male SD rats were assigned to four groups: control (CTR), finasteride 1 mg/kg/day (F), finasteride 1 mg/kg + DA-9401 100 mg/kg/day (F + DA 100) and finasteride 1 mg/kg + DA-9401 200 mg/kg/day (F + DA 200). Treatments were by oral delivery once daily for 90 consecutive days. The gross anatomical parameters assessed included: genital organ weight; vas deferens sperm count and sperm motility; testosterone, dihydrotestosterone (DHT) and malondialdehyde levels; and histological and terminal deoxynucleotidyl transferase enzyme mediated dUTP nick-end labeling (TUNEL) staining of testis for spermatogenic cell density, Johnsen's score and apoptosis. Testicular tissue was also used for evaluating endoplasmic reticulum (ER) stress and apoptotic proteins. Epididymis weight, seminal vesicle weight, prostate weight, penile weight and vas deferens sperm motility showed significant differences between the F group and the CTR, F + DA 100 and F + DA 200 groups. There was no significant change in the testosterone level. DHT level decreased significantly in the F group compared with the CTR group. Testis tissue revealed significant changes in spermatogenic cell density, Johnsen's score and apoptotic index. Western blot showed significant changes in the ER stress and apoptotic markers. Finasteride resulted in reduced fertility and increased ER stress and apoptotic markers, which were recovered by administration of DA-9401 in the SD rats.

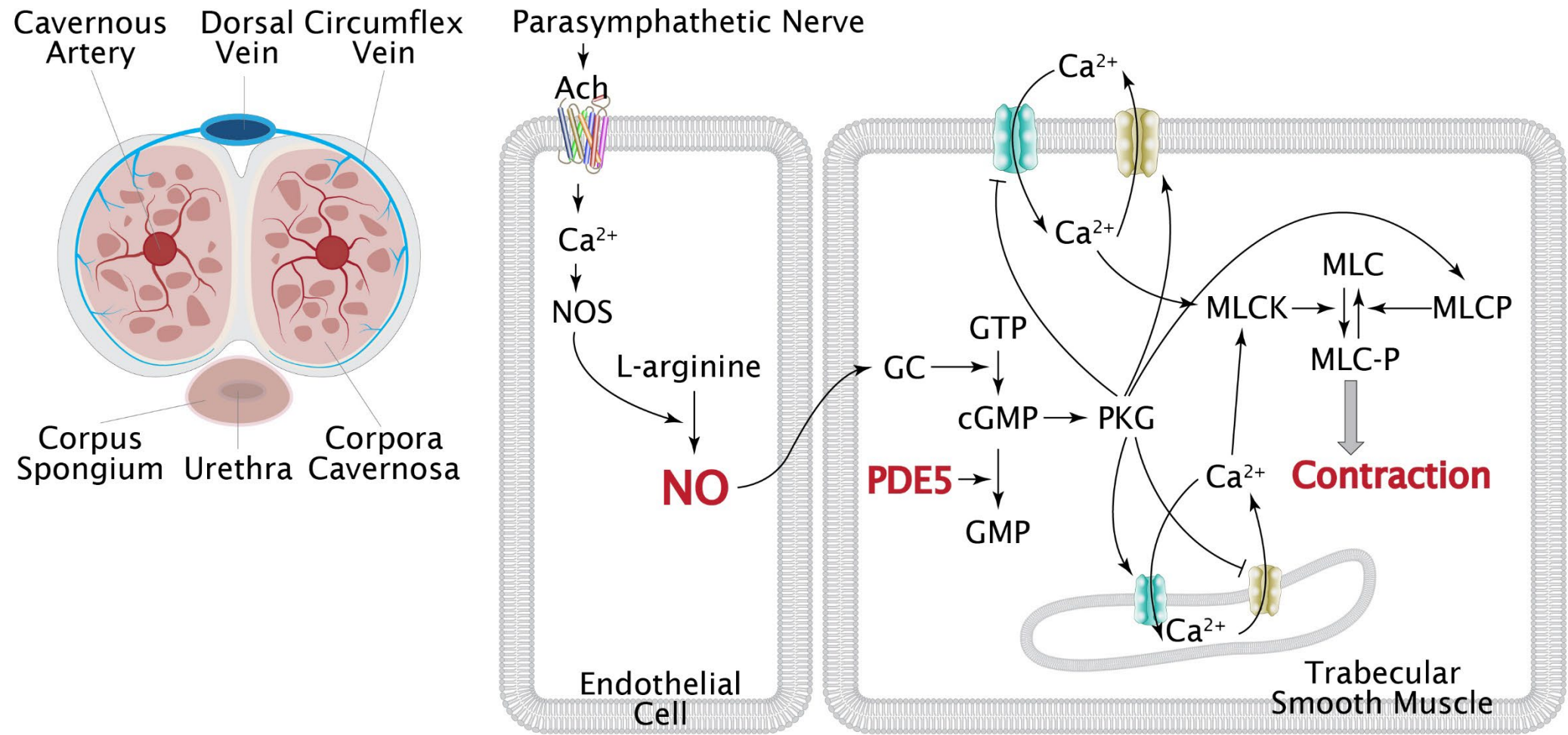
Keywords: finasteride, DA-9401, infertility, sperm, dihydrotestosterone, endoplasmic reticulum stress, apoptosis

Drug Design, Development and Therapy 2017;11:2969-2979

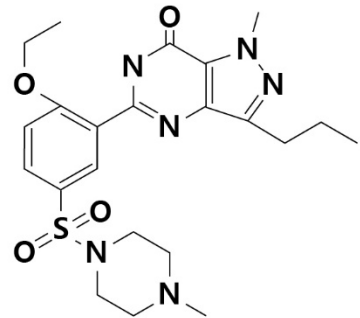
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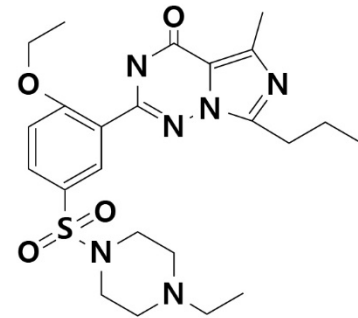
► Mechanism of penile erection



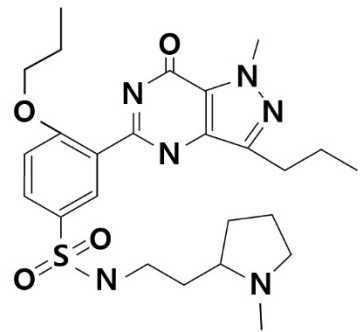
► Drugs on erectile dysfunction



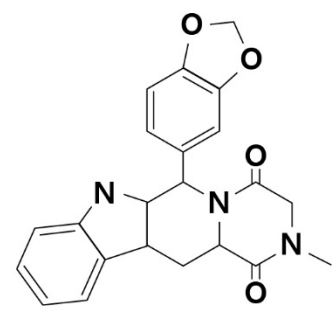
sildenafil



verdenafil



eudenafil



tadalafil

Drugs	Therapeutic Uses
PDE5 inhibitors	
Sildenafil	Erectile Dysfunction
Vardenafil	Pulmonary Arterial Hypertension
Tadalafil	(Sildenafil, Tadalafil)
Avanafil	

► Oral contraception

제제	에스트로젠(mg)	프로제스틴(mg)
경구 복합 피임제		
1상 제제		
ethinyl estradiol/norethindrone	1차 0.02	1차 1.0
	0.03	1.5
	0.035	0.5
	0.035	1.0
	0.035	0.4
	0.05	1.0
ethinyl estradiol/levonorgestrel	0.03	0.15
ethinyl estradiol/norgestrel	0.03	0.3
	0.05	0.5
ethinyl estradiol/ethynodiol diacetate	0.035	1.0
	0.05	1.0
ethinyl estradiol/desogestrel	0.03	0.15
ethinyl estradiol/norgestimate	0.035	0.25
mestranol/norethindrone	0.05	1.0

제제	에스트로젠(mg)	프로제스틴(mg)
2상 제제		
ethinyl estradiol/norethindrone	1차/2차 0.035 ^{10일} /0.035 ^{11일} 0.035 ^{7일} /0.035 ^{14일}	1차/2차 0.5/1.0 0.5/1.0
3상 제제		
ethinyl estradiol/norethindrone	1차/2차/3차 0.035/0.035/0.035 0.035 ^{7일} /0.035 ^{9일} /0.035 ^{7일}	1차/2차/3차 0.5/0.75/1.0 0.5/1.5/0.5
ethinyl estradiol/norgestimate	0.035/0.035/0.035 0.025/0.025/0.025	0.18/0.215/0.25 0.18/0.215/0.25
ethinyl estradiol/levonorgestrel	0.036 ^{6일} /0.04 ^{5일} /0.03 ^{10일} 0.03 ^{6일} /0.04 ^{5일} /0.03 ^{10일}	0.05/0.075/0.125 0.05/0.075/0.125
프로제스틴 단독 피임제		
norethindrone	-	0.35
norgestrel	-	0.075

※ 1상 제제는 21일 제제이고 3상 제제 중 별도 표기가 없는 제제는 7일/7일/7일 제제임

► Summary of female hormone and drugs

Drugs	Therapeutic Uses
Estrogens	
Steroidal	
Estradiol	Menopause hormone therapy Oral contraceptive
Ethinyl estradiol	
Mestranol	
Equilin	
Estron sulfate	
Nonsteroidal	
Diehtylstilbestrol	
Bisphenol A	
Genistein	
Estrogen Receptor Modulator	
Tamoxifen	Breast caner
Raloxifene	Osteoporosis
Toremifene	Breast cancer
Antiestrogen	
Clomiphene	Ovulation
Fulvestrant	Breast cancer resistant to tamoxifen

Drugs	Therapeutic Uses
Estrogen Synthesis Inhibitor	
Steroidal	Breast cancer
Formestane	
Exemestane	
Nonsteroidal	
Anastrozole	
Letrozole	
Vorozole	
Progestins	
Pregnanes	Menopause hormone therapy Contraception
Progesterone	
Medroxyprogesterone	
Megestrol	
Estranes	Contraceptive
Norethindrone	
19-norethindrone	
Gonanes	Contraceptive
Norgestrel	
Norgestimate	

Drugs	Therapeutic Uses
Antiprogesterone and modulator	
Antagonist	
Mifepristone	Termination of pregnancy
Partial agonist	
Ulipristal	Emergency contraception

► Summary of male hormone and drugs

Drugs	Therapeutic Uses
GnRH Analogs	
Leuprolide	Metastatic Prostate Cancer
Goserelin	Endometriosis(Leuprolide)
Triptorelin	Breast Cancer(Goserelin)
Histrelin	
Buserelin	
AR Antagonists	
Flutamide	Metastatic Prostate Cancer
Bicalutamide	
Nilutamide	
Enzalutamide	

Drugs	Therapeutic Uses
5α-Reductase Inhibitors	
Finasteride	Benign Prostate Hyperplasia
Dutasteride	Alopecia(Finasteride)
PDE5 inhibitors	
Sildenafil	Erectile Dysfunction
Vardenafil	Pulmonary Arterial Hypertension
Tadalafil	(Sildenafil, Tadalafil)
Avanafil	